Research activity

A reservoir topped up every day — 35 min.

A reservoir initially contains 100 litres of water. Each day, 20% of the volume present is drawn off, then 30 litres are added. For every natural number n, let u_n be the volume of water, in litres, after these two operations have been carried out n times. Thus $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} in terms of u_n.
2. Which volume would remain unchanged after one day? Prove by induction that, for every $n \in \mathbb{N}$, $100 \leq u_n < 150$.
3. For every $n \in \mathbb{N}$, let $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions of v_n and u_n in terms of n.
4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.
5. A student has calculated the first hundred terms and notes that they are all less than 150. Is this observation enough to prove that all the terms are? Justify.